
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has become the central lever for eliciting reasoning
2 in large language models, yet the mechanism by which RL reshapes a model
3 over the course of training has remained largely uncharacterized. We close this
4 gap with a single, unified account of the RL training process, one that couples a
5 systematic empirical dissection of the training trajectory with a rigorous theoretical
6 characterization of its dynamics. Empirically, jointly resolving the trajectory at
7 the reasoning-pattern level and the token level yields a coherent and mutually
8 reinforcing picture: the intrinsic success rate of each individual reasoning pattern
9 remains essentially invariant throughout training, whereas RL concentrates its opti-
10 mization on a sparse subset of critical tokens and, acting through them, *reshapes*
11 *the distribution over reasoning patterns* to produce the observed changes in per-
12 formance. Building directly on this evidence, we develop a theoretical framework
13 that governs the training dynamics of RL under both prevailing reward regimes—
14 reinforcement learning with verifiable rewards (RLVR) and reinforcement learning
15 with internal feedback (RLIF)—within one formalism. For RLVR, we characterize
16 two canonical regimes, one in which the model converges rapidly to the optimal
17 reasoning strategy and one in which an intervening entanglement stage provably
18 retards optimization, and we identify the reasoning quality of the base model as the
19 decisive factor governing which regime obtains. For RLIF, we establish precisely
20 why internal rewards raise performance in the early phase of training and why
21 continued optimization can nonetheless drive the model strictly below its starting
22 point, deriving both the benefit and the characteristic failure mode as two phases
23 of a single trajectory. Extensive experiments spanning multiple RL algorithms,
24 base models across a range of scales, and mathematical reasoning benchmarks
25 corroborate every component of the analysis, together advancing the theoretical
26 understanding of RL post-training and yielding concrete, actionable guidance for
27 strengthening the reasoning capabilities of language models.

28 1 Introduction

29 State-of-the-art reasoning models such as Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1 [14]
30 now attain exceptional performance across a broad spectrum of demanding logical tasks, spanning
31 mathematics [30, 44, 41] and programming [49, 40, 8]. Reinforcement learning (RL) is the central
32 technique underpinning this progress, reliably and reproducibly elevating model reasoning to a
33 qualitatively higher level. Understanding *how* RL achieves this—the mechanism by which it converts
34 a base model into a capable reasoner—is therefore of first-order importance for the principled design
35 of post-training, and it is the question this work resolves.

36 This progress has motivated a growing body of work on the mechanisms of RL for LLMs, and each
37 existing line captures a genuine but partial facet of the phenomenon. Comparing pass@k performance,

Yue et al. [42] observe that RL-post-trained models rarely surpass their base models, indicating that RL does not elicit fundamentally new reasoning patterns but rather reweights those the model already possesses. From an entropy standpoint, Cui et al. [9], Zhang et al. [45] prove that RL-based methods reduce policy entropy, and Cui et al. [9] further establish an empirical link between policy entropy and model performance. Complementing these results, Wang et al. [33], Huan et al. [17] show that RL concentrates its updates on a sparse subset of critical tokens. Each of these lines, however, isolates a single symptom in isolation; what has remained missing—and what we supply—is a unified account of the RL training *dynamics* that binds these facets into one coherent mechanism and grounds that mechanism in both direct empirical evidence and formal analysis.

We characterize the RL training process through a systematic analysis that jointly resolves the reasoning-pattern level and the token level across the entire training trajectory. Concretely, we track next-token probability-rank shifts throughout training and, in parallel, apply complementary LLM-based and rule-based procedures to extract and classify the reasoning patterns realized in model responses, quantifying how their per-pattern success rates and selection frequencies co-evolve. This joint, multi-resolution design yields clearer and more decisive evidence than prior single-level analyses [17, 42]: RL optimizes only a sparse subset of critical tokens and, acting through them, *reshapes the distribution over reasoning patterns* to determine model performance, while the intrinsic success rate of each individual pattern remains essentially invariant throughout training. These two findings—sparse token-level intervention that drives a pattern-level reallocation of probability mass, atop fixed per-pattern success rates—are precisely the structural ingredients that both motivate and constrain the mathematical framework we develop next.

Guided by these findings, we develop a theoretical framework that formalizes reasoning as a two-stage question–reason–answer process $q \rightarrow r \rightarrow a$: the model first selects a reasoning pattern conditioned on the question, $\pi(r|q)$, and then generates a final answer conditioned on that pattern, $\pi(a|r, q)$, with RL acting principally on the first, pattern-selection stage. This decomposition translates our empirical observations directly into provable statements, and it applies uniformly to both dominant reward types—verifiable rewards (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we prove that optimization converges to the reasoning pattern r^* with the highest success rate—exactly the behavior our experiments exhibit—and we further delineate two qualitatively distinct convergence regimes: models with strong initial reasoning quality converge rapidly to the optimal pattern, whereas weaker models must first traverse an entanglement stage in which a suboptimal pattern provably retards optimization. For RLIF, we prove both why internal rewards improve performance early in training and why continued optimization can nonetheless converge to a state strictly worse than the base model, again matching what we observe empirically. Dedicated case studies and numerical simulations confirm the theory across both reward regimes.

Our main contributions are as follows.

- **A unified empirical characterization of RL training dynamics.** We conduct a systematic, multi-resolution analysis that jointly resolves the reasoning-pattern level and the token level across the full RL training trajectory. By coupling next-token probability-rank shifts with complementary LLM-based and rule-based pattern analysis, we obtain clearer and more decisive evidence than prior single-level work [17, 42], establishing that RL optimizes only a sparse subset of critical tokens and thereby *reshapes the distribution over reasoning patterns* to determine performance, while the intrinsic success rate of each individual pattern remains essentially invariant during training.
- **A theoretical framework with provable dynamics for both reward regimes.** We introduce a formal two-stage model of reasoning, $q \rightarrow r \rightarrow a$ (question–reason–answer), and use it to characterize the training dynamics of both dominant RL paradigms within a single account. For RLVR, we prove convergence to the highest-success-rate reasoning pattern and rigorously separate two regimes—rapid convergence for strong base models versus a provably slow entanglement stage for weaker ones—thereby identifying base-model reasoning quality as the decisive determinant of convergence. For RLIF, we prove that early-stage performance gains are guaranteed at initialization yet that continued training can converge to a state strictly worse than the base model, explaining both the benefit and the characteristic failure mode of internal-feedback rewards from first principles.
- **Empirical validation and practical guidance.** We corroborate the framework through case studies and numerical simulations that quantitatively reproduce both RLVR regimes and RLIF’s non-monotone rise-then-collapse behavior. In doing so, we close the gap between empirical observation

94 and theoretical understanding of RL training dynamics and distill concrete, actionable guidance for
 95 designing and improving RL-based LLM post-training.

96 2 Related Works

97 **Reinforcement Learning for LLMs** Reinforcement learning has become a cornerstone technique for
 98 advancing large language models (LLMs), both in aligning model behavior with human preferences
 99 [25, 48] and in solving complex mathematical and programming tasks [30, 20, 22]. The reward
 100 model sits at the center of this pipeline and has classically been trained on extensively curated, human-
 101 annotated data [26, 25, 2]. A decisive recent shift has supplanted such costly supervision with rewards
 102 that are cheaply obtainable and automatically verifiable, as instantiated by reinforcement learning
 103 with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF)
 104 [47, 4]. This shift has, in turn, driven a wave of new optimization algorithms, including GRPO
 105 [30], DAPO [41], and VAPO [43]. By dispensing with a separately trained reward model, these two
 106 regimes constitute the dominant paradigms for reasoning-oriented post-training—and are precisely
 107 the setting our analysis targets.

108 **Theoretical Analyses and Mechanism Interpretation of RL for LLMs.** The empirical success of
 109 RL for LLMs has spurred extensive work on theoretical analysis and mechanism interpretation. Early
 110 efforts operated largely at a high level, studying how to design better rewards [28, 18, 38] and how
 111 to construct novel training algorithms [37, 11, 21]. As the field moved toward the simpler reward
 112 structures of RLVR and RLIF, attention shifted to explaining *why* these paradigms are so effective,
 113 and several complementary accounts have emerged: Shao et al. [29] show that even spurious rewards
 114 can enhance reasoning by unlocking latent skills acquired during pretraining; Agarwal et al. [4]
 115 attribute the gains to entropy minimization; Gandhi et al. [13] isolate specific cognitive behaviors,
 116 such as backtracking, that improve reasoning; and Zhao et al. [46] demonstrate that RL fine-tuning
 117 amplifies pre-trained behaviors and converges toward a dominant output format, a phenomenon that
 118 Wang et al. [34], Cui et al. [10] further dissect through entropy collapse. Each of these perspectives,
 119 however, illuminates a single symptom of RL training in isolation. We instead analyze the RL training
 120 *dynamics* through the unifying lens of reasoning-pattern selection and, to the best of our knowledge,
 121 are the first to substantiate this view with a complete mathematical framework and accompanying
 122 convergence guarantees that jointly account for both RLVR and RLIF.

123 3 Preliminaries

124 **Reinforcement Learning for LLMs** Let π_{θ} denote a language model with parameters θ , treated as the
 125 policy to be optimized. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy π_{θ} autoregressively
 126 generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$. RL optimizes the KL-regularized expected-reward
 127 objective

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\cdot | \mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

128 where π_{ref} is the base reference policy and the hyperparameter β weights the KL-divergence penalty
 129 that constrains π_{θ} from drifting away from π_{ref} .

130 The reward $r_{\phi}(\mathbf{x}, \mathbf{y})$ admits several instantiations, ranging from a separately trained reward model
 131 [25] to a rule-based scoring function [30]. We concentrate on the two verification-light regimes that
 132 now dominate reasoning-oriented post-training: verifiable rewards (RLVR) [14, 30] and the model’s
 133 internal feedback (RLIF) [47, 4]. The two regimes differ precisely in the provenance of the reward
 134 signal, and this single distinction proves decisive for the divergent training dynamics we analyze.

135 Under RLVR, the reward $r_{\phi}(\mathbf{x}, \mathbf{y})$ evaluates directly whether the generated answer $\mathbf{y}$ matches the
 136 ground-truth answer to $\mathbf{x}$, yielding the binary verifiable criterion

$$r_{\phi}(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

137 In contrast, RLIF [47, 4] dispenses entirely with any external verifier and derives its reward solely from
 138 intrinsic model signals. Following Zhao et al. [47], we instantiate the RLIF reward as the negative
 139 average KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s

next-token distribution, so that a higher reward corresponds directly to higher model confidence:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U || \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

Together, RLVR and RLIF span the two prevailing reward regimes for reasoning-oriented post-training. Our objective is to characterize the training dynamics of both—empirically and theoretically—within a single unified framework, thereby explaining their divergent behavior from common principles.

4 Experimental Exploration for RL Training

To characterize the RL training process, we conduct a systematic, multi-resolution study of RL under both reward types, organized so that each level of analysis accounts for the one above it. We begin with a high-level overview of the training procedure, which establishes a decisive behavioral contrast: RLVR yields steady, monotone improvement, whereas RLIF is unstable and can ultimately degrade performance below that of the base model. We then descend to a reasoning-pattern-level analysis, which localizes the cause of this contrast—RLVR progressively concentrates probability on reasoning patterns with higher success rates, whereas RLIF fails to specialize toward the effective ones. Finally, a token-level analysis exposes the underlying mechanism: these pattern-level reallocations are implemented by rank changes in a strikingly small fraction of tokens. Taken together, and in direct comparison with prior single-level work [17, 42], these three levels of analysis furnish clearer and more decisive evidence that RL optimizes only a sparse subset of critical tokens and, acting through them, *reshapes the distribution over reasoning patterns* to determine model performance.

4.1 Training Procedure Overview for RLVR and RLIF

We first examine the complete training trajectories of RLVR and RLIF under a controlled comparison. Taking Qwen2.5-3B [39] as the base model, we train it on the MATH dataset [16] while holding every other setting fixed across the two algorithms, so that the reward is the sole source of variation; multiple runs per paradigm are obtained by varying the random seed.

As Figure 1a shows, the two methods trace sharply divergent performance trajectories. RLVR induces a stable optimization process in which accuracy improves steadily and converges smoothly over training. RLIF, by contrast, raises performance only transiently: after early gains, continued training drives accuracy back down, in some runs below that of the original base model. This qualitative divergence—monotone improvement under RLVR versus a rise-then-collapse trajectory under RLIF—is precisely the phenomenon our subsequent fine-grained analysis is designed to explain, and it recurs consistently across seeds.

4.2 Reasoning-pattern Level Analysis

To pinpoint the source of the performance disparity between RLVR and RLIF, we carry out a fine-grained analysis at the reasoning-pattern level. To define and extract patterns, we first collect responses from the base model and employ GPT-4o [19] to cluster them into distinct categories according to their keywords and logical structure, generating a descriptive summary for each category. Equipped with this classifier, we then jointly track the selection frequency of each reasoning pattern together with its associated per-pattern success rate across the entire RL training procedure, so that any change in overall accuracy can be attributed directly to its pattern-level origin rather than to opaque parameter drift.

Concretely, we trace the joint dynamics of pattern selection and per-pattern accuracy under both RLVR and RLIF. Figure 1b reports this analysis on Number Theory tasks drawn from the MATH dataset [16], and the same behavior reproduces consistently across additional models and datasets (deferred to Appendix A). Three findings emerge, and each holds uniformly across every setting we examine:

- RLVR-trained models shift monotonically toward reasoning patterns with higher success rates, which directly and quantitatively accounts for the steady improvement in overall accuracy under RLVR.

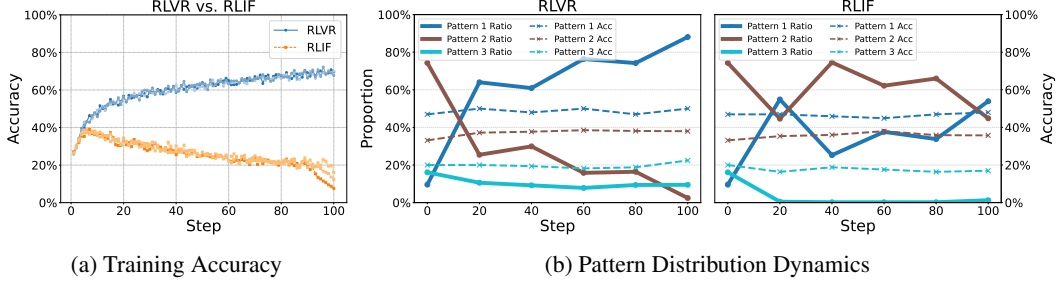


Figure 1: **(a)** Training trajectories of RLVR and RLIF on the MATH dataset (Qwen2.5-3B base model), with multiple runs per paradigm obtained by varying the random seed. The contrast is decisive: RLVR delivers stable, monotone gains, whereas RLIF drives accuracy up and then back down. **(b)** The reasoning-pattern-level view of the same training runs. Solid lines report each pattern’s selection frequency among all patterns (left axis) and dotted lines report its corresponding per-pattern accuracy (right axis). Under RLVR the model progressively concentrates probability on the higher-accuracy patterns while abandoning the lower-accuracy ones; under RLIF the pattern distribution evolves erratically and never specializes. Patterns are sorted and named by decreasing accuracy for readability, so that a selection shift toward Pattern 1 maps directly onto a gain in overall accuracy—rendering the pattern-level origin of the accuracy trajectories in (a) immediately legible.

- RLIF, in contrast, induces an unstable pattern distribution that never specializes toward the more effective patterns, precisely mirroring its erratic accuracy trajectory.
- Under both methods, the success rate of each individual reasoning pattern remains stable throughout training, so that performance changes are attributable to *which* pattern is selected rather than to any change within a pattern.

Together these results localize the RLVR–RLIF performance gap squarely in the divergent pattern-selection dynamics of the two rewards: RLVR reallocates probability toward high-success patterns whereas RLIF does not, even though the patterns themselves are equally effective under both. This firmly locates the operative mechanism at the level of pattern selection and directly motivates the token-level analysis that follows, which asks what, inside the model, actually implements these reallocations.

4.3 Token-level Analysis

To uncover the mechanism that implements the reasoning-pattern dynamics identified above, and building on Huan et al. [17], Yue et al. [42], we conduct a complementary token-level analysis. We sample base-model responses to questions from GSM8K [6], MATH [16], and AIME [7], and then, holding these responses fixed, contrast the rank each token receives under the base model with its rank under the RL-enhanced model, where a token’s rank is the position of its probability among all vocabulary tokens given the preceding context. As summarized in Table 1, the ranks are overwhelmingly stable: fewer than 10% of tokens change rank in essentially every combination of task, method, and training step. RL therefore operates as a sparse intervention, adjusting the probabilities of a small set of critical tokens while leaving the overwhelming majority of the reasoning trajectory untouched.

Table 1: Fraction of tokens whose probability rank shifts after RL, broken down by task, method, and training step. Across GSM8K, MATH, and AIME24, and for both RLVR and RLIF, only a small minority of tokens change rank—below 10% in every single case—confirming that RL intervenes sparsely at the token level, and that this sparsity is invariant to task, reward regime, and training step.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

Taken together, the three levels of analysis converge on a single, clean characterization of RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning*

218 *pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual*
 219 *patterns remains relatively stable during training.*

220 Beyond deepening the empirical understanding of RL post-training, this characterization supplies
 221 the two structural ingredients—sparse token-level control of pattern selection, atop fixed per-pattern
 222 success rates—on which we build the mathematical framework and formal analysis of the next
 223 section, thereby elevating these empirical regularities into provable statements.

224 5 Theoretical Characterization of RL Training Dynamics

225 We now move from empirical characterization to formal explanation of the RL training process.
 226 Building directly on the two structural findings of the previous section, we abstract reasoning as a
 227 two-stage procedure—first selecting a reasoning pattern, then deducing an answer conditioned on
 228 it—in which RL principally optimizes the selection stage. This framework converts our empirical
 229 observations into provable statements about RL under both reward regimes. For RLVR, we analyze
 230 the full training dynamics and rigorously separate two regimes exhibiting qualitatively distinct
 231 optimization behavior, tracing the separation to the reasoning quality of the base model. For RLIF,
 232 we prove both why internal rewards improve performance at the outset of training and, from the very
 233 same analysis, why continued optimization can converge to a model strictly worse than the base
 234 model.

235 5.1 A Theoretical Framework for Reasoning Models

236 **Abstract Reasoning Process.** We formalize reasoning as the following two-stage generative
 237 process: given a question q , the model (1) selects a reasoning pattern r from the candidate set
 238 $\mathcal{R} = \{r_1, r_2, \dots\}$ and (2) generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ conditioned on that
 239 pattern. Pattern r_i is selected with probability $p(r_i|q)$, and each pattern carries its own success
 240 rate—the probability that it yields the correct answer. This decomposition lets us recast the RLIF
 241 objective (Eq. 3.3) directly in terms of the pattern-selection and answer distributions, thereby exposing
 242 the pattern-selection stage on which RL principally acts¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

243 **Policy Parameterization.** Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$
 244 the set of output sequences. Given a question q , the language model defines a distribution over output
 245 sequences $y \in \mathcal{Y}$ autoregressively via

$$\text{(general policy)} \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

246 where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is parameterized by θ and predicts the l -th token from the preceding context
 247 $y_{<l}$, and the next token follows $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ with $z = f_\theta(x, y_{<l})$.

248 Because the full expressive power of practical models renders end-to-end optimization guarantees
 249 analytically intractable [3, 23], we follow a well-established line of theoretical work [27, 24, 10, 45]
 250 and adopt a *tabular policy* parameterization. This is the special case of Eq. 5.2 in which each output
 251 is assigned its own trainable logit indexed by the preceding token; concretely, for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$,

$$\text{(tabular policy)} \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

252 where $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ associated with the preceding token y_{l-1} . This parame-
 253 terization retains the softmax structure of the general policy while rendering the training dynamics
 254 analytically tractable, and thereby affords provable statements that would otherwise lie out of reach.

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

255 **Optimization for RL.** To analyze the training dynamics, we combine the tabular policy of Eq.5.3
 256 with the RL objective $\phi_{\text{RL}}(\theta)$ of Eq.3.1 and study the policy gradient in the small-learning-rate limit,
 257 that is, the gradient flow

$$\frac{d}{dt}\theta(t) = \nabla\phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

258 where $\theta(t)$ denotes the parameters of the policy $\pi_{\theta(t)}$ at training time t , initialized at $\theta(0) = \theta_{\text{ref}}$.

259 Our experiments established that the success rate of each individual reasoning pattern stays essentially
 260 fixed throughout training. We attribute this invariance to an inherent asymmetry in optimization
 261 difficulty: reshaping the mapping from questions to reasoning patterns is markedly easier than altering
 262 the mapping from reasoning patterns to final answers. We formalize this empirically grounded
 263 observation as the following assumption, which underpins the analysis throughout this section.

264 **Assumption 5.1.** *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer*
 265 *$\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

266 Throughout the remainder of this section we adopt Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$
 267 for the fixed success rate of pattern $\mathbf{r}$. We proceed in two steps: we first derive the optimal policy
 268 for the RL objective (Eq.3.1) under the general autoregressive policy (Eq.5.2), which identifies the
 269 *destination* of optimization; we then characterize the full RLVR and RLIF training dynamics under
 270 the tabular policy (Eq. 5.3), which identifies the *path* taken to reach it.

271 5.2 The Optimal Policy for RL

272 Our experiments showed that RLVR improves performance steadily by promoting higher-success-rate
 273 reasoning patterns, whereas RLIF yields only unstable, non-monotone gains. We now trace this
 274 contrast to its root by characterizing the optimal policy toward which each objective drives the model.

275 **Proposition 5.2.** *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive*
 276 *policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

277 where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the
 278 reasoning path reward. Specifically:

- 279 • For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- 280 • for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution,
 281 i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

282 Proposition 5.2 pinpoints the optimal solutions for RLVR and RLIF and, in doing so, explains their
 283 divergent behavior at its source. Assuming π_{opt} lies within the general autoregressive policy space—
 284 which holds for both practical LLMs and the tabular policies analyzed below—the RLVR-optimal
 285 policy up-weights a reasoning pattern $\mathbf{r}$ precisely when its probability of deducing the correct answer
 286 is high enough that $\exp(\frac{1}{\beta}p^*(\mathbf{r}))/Z > 1$; the reward is thus *aligned with correctness*. The RLIF
 287 reward $R_{\text{RLIF}}(\mathbf{r})$, by contrast, depends only on the confidence of the final-answer distribution and
 288 therefore cannot distinguish the correct answer $\mathbf{a}^*$ from confidently held incorrect alternatives. This
 289 single structural difference is decisive: the RLVR optimum reliably improves accuracy, whereas the
 290 RLIF optimum carries no such guarantee and may instead favor patterns that are confident yet wrong.

291 Moreover, since β is typically small in practice (e.g., 0.001) [26, 5, 44], taking the limit $\beta \rightarrow 0$
 292 sharpens this picture into a deterministic selection rule:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

293 In this limit the policy collapses onto the single reasoning pattern with the largest reward $R(\mathbf{r})$,
 294 independently of how the reference model initially distributes probability across patterns. This
 295 characterizes the *destination* of optimization, but not the *path* to it: because the objective landscape is
 296 non-convex, the optimum alone reveals neither how reliably RLVR reaches high-reward patterns nor
 297 why RLIF can improve performance before deteriorating. We resolve both questions in the next two
 298 subsections by analyzing the full training dynamics under the tabular policy (Eq. 5.3).

5.3 Training Dynamic Analysis for RLVR

Our experiments established that the magnitude of RLVR’s gains depends strongly on the capacity of the base model. We now explain this dependence analytically by tracking, under gradient flow, the probability assigned to the *optimal reasoning pattern*—the pattern $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$ with the highest success rate for reaching the correct answer $\mathbf{a}^*$. The dynamics split cleanly into two regimes. In the first, $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically and converges to 1. In the second, the model must first traverse an *entanglement stage* in which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ absorbs probability mass and stalls the growth of this selection probability; only after this stage does the model transition into the first-regime dynamics and ultimately converge to $\mathbf{r}^*$. The following two theorems render each regime, and the boundary between them, precise.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})p^*(\mathbf{r})$, satisfies:*

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 isolates the favorable regime: the base reference model is strong enough that its overall accuracy already exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). Under this condition RLVR converges to the optimal pattern at the fast rate $\mathcal{O}(1/\epsilon)$, so that the model rapidly attains an overall accuracy approaching the success rate of $\mathbf{r}^*$.

The complementary—and empirically well-documented—regime arises when the base model is weaker, a setting in which RLVR optimization is known to be difficult [44, 36]. Here the model first enters an entanglement stage in which the *second-best pattern*, $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, competes for probability mass and thereby retards the optimization of $\mathbf{r}^*$. Theorem 5.4 quantifies precisely how long this stage persists.

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(\text{Regime 2}) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.

Theorem 5.4 analyzes the representative case in which only the optimal and second-best patterns have success rates exceeding the overall accuracy. We adopt this setting for analytical clarity; the argument extends, at the cost of heavier bookkeeping, to the more general situation in which *at least* two patterns exceed the average success rate.

The central consequence of Theorem 5.4 is a sharp separation of time scales. Reaching the favorable regime of Theorem 5.3—where the overall accuracy surpasses the success rate of the second-best pattern and the optimal pattern is selected with substantial probability—requires T_0 steps. Whereas the subsequent convergence time T_1 of Theorem 5.3 is merely polynomial in $1/\epsilon$, the transition time T_0 can grow *super-exponentially* in $\gamma_{\pi_{\text{ref}}}$, the ratio between the total selection-weighted success mass carried by $\mathcal{R}/\{\mathbf{r}'\}$ and that carried by $\mathbf{r}^*$. When the base model places only vanishing probability on the optimal pattern, $\gamma_{\pi_{\text{ref}}}$ grows large and T_0 becomes prohibitive, so the model remains stuck for an extended period before it begins to select $\mathbf{r}^*$ with appreciable probability—this is precisely the entanglement stage. The result thus furnishes a rigorous explanation for why RLVR succeeds almost effortlessly on strong base models yet stalls on weak ones, and it identifies base-model reasoning quality as the decisive determinant of convergence.

We corroborate both regimes with case studies and numerical simulations in Section 6.

5.4 Theoretical Explanation for RLIF

Proposition 5.2 already establishes that RLIF, being driven solely by the confidence of the final-answer distribution, cannot distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives, and hence that its optimum may lie below the base model in accuracy—consistent with the degradation we observe empirically. Yet prior work reports that RLIF can improve performance without any external reward Agarwal et al. [4], Zhao et al. [47]. We reconcile these seemingly opposing facts by analyzing RLIF’s full dynamics, establishing that both the early improvement and the eventual collapse follow from a single theory as two phases of one trajectory.

We begin with the early stage, considering a well-trained base model that satisfies the following assumption.

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.*

Because majority voting [35] is an established means of improving model performance, a well-trained LLM typically assigns the correct answer $\mathbf{a}^*$ higher probability than any incorrect one, so Assumption 5.5 is well justified for modern LLMs. Under this assumption we analyze the accuracy dynamics at initialization ($t = 0$) in the case $|\mathcal{A}| = 2$, with per-pattern success rates drawn uniformly and a high-entropy pattern-selection distribution², which yields the following theorem.

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. *The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.*
2. *With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.*

The two results of Theorem 5.6 together resolve the apparent paradox surrounding RLIF. The first shows that the accuracy derivative at initialization is positive with probability one, which precisely explains RLIF’s early performance gains. The second exposes the failure mode that inevitably follows: as training proceeds, Proposition 5.2 implies that at $\beta = 0$ the policy converges deterministically to the pattern maximizing $R_{\text{RLIF}}(\mathbf{r})$, and with probability 0.5 this maximizer coincides with the least accurate pattern, $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. RLIF therefore improves the model at the outset yet, with probability one-half, ultimately converges toward the lowest-accuracy reasoning path, driving performance below that of the base model. This rise-then-collapse trajectory is exactly the behavior we characterize theoretically here and observe empirically—now derived from first principles rather than merely reported.

6 Case Studies and Numerical Simulations

To validate and concretely illustrate the theory, we present case studies and training simulations that instantiate each theorem. For RLVR, we realize both regimes of Theorems 5.3 and 5.4, shown in Fig. 2 (upper). For RLIF, we run repeated training simulations in which each reasoning path’s success rate is drawn uniformly, directly probing RLIF’s behavior both at the initial training steps and at convergence, shown in Fig. 2 (bottom). In every case the simulated dynamics match the theoretical predictions quantitatively, confirming that the framework captures the observed behavior rather than merely being consistent with it.

6.1 Case Studies for RLVR

Section 5.3 established two regimes in RLVR’s training dynamics: in the first (Eq 5.7), the probability $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ of selecting the optimal pattern rises monotonically to convergence; in the second (Eq 5.8), an initial entanglement phase imposes a substantially longer training period before the first-regime dynamics take over and the model converges to $\mathbf{r}^*$. The case studies confirm both regimes precisely:

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest-entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

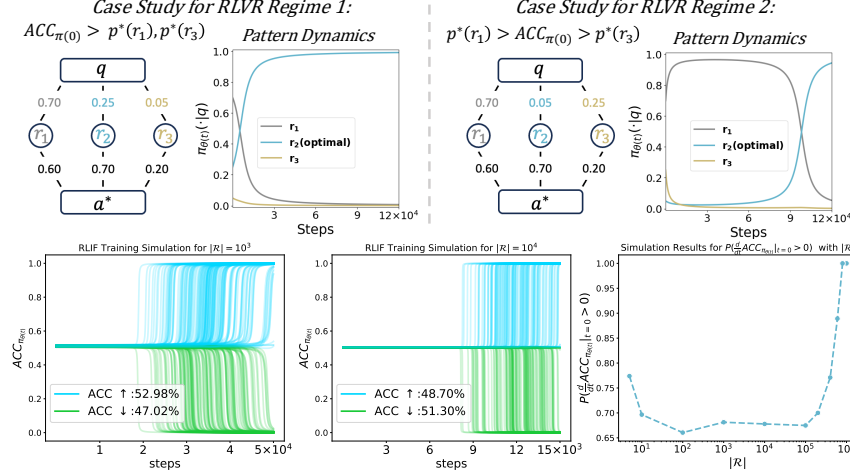


Figure 2: Empirical validation of the theory. *Upper (RLVR case studies)*: the two regimes of Theorems 5.3 and 5.4—rapid convergence to the optimal pattern (left) and the entanglement stage that delays it (right). *Bottom (RLIF simulations)*: as $|\mathcal{R}|$ varies, the probability of converging to an improved versus a degraded state (left) and the probability of an initial performance gain (right). The simulations reproduce both predictions of Theorem 5.6 quantitatively: near-certain early improvement together with a near one-half chance of eventual degradation, in close agreement with the theoretical values.

- When the reference model satisfies Eq 5.7—its overall accuracy exceeding the success rate of every non-optimal pattern—the model converges rapidly to the optimal reasoning pattern, as shown in Fig. 2 (upper left).
- When instead a suboptimal pattern r' carries a higher success rate than the initial overall accuracy (Theorem 5.4), the model first traverses the entanglement phase, and the transition period T_0 (Theorem 5.4) must elapse before the rapid first-regime optimization begins. Fig. 2 (upper right) instantiates a large- $\gamma_{\pi_{\text{ref}}}$ scenario (with $\gamma_{\pi_{\text{ref}}}$ defined in Eq 5.9, here $\gamma_{\pi_{\text{ref}}} = 6$, since $(0.25 + 0.05)/0.05 = 6$); because T_0 scales as $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$, the transition becomes prohibitively long and convergence to the optimal pattern is markedly delayed. The observed pattern dynamics exhibit exactly this entanglement stage, in close quantitative agreement with the theory.

6.2 Numerical Simulations For RLIF

Section 5.4 established that, under Assumption 5.5 and the base-model distribution of Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, while the probability of converging to the least accurate reasoning path is exactly 0.5. Our simulations confirm both predictions quantitatively. Fig. 2 (bottom left) reports RLIF training simulations across varying $|\mathcal{R}|$: the probability of converging to an improved state ($ACC \uparrow$) closely tracks that of converging to a degraded state ($ACC \downarrow$), matching the second result of Theorem 5.6. Fig. 2 (bottom right) further shows that, at the initial training step, the probability of performance improvement approaches 1 as $|\mathcal{R}|$ grows, matching the first result. The simulations thus reproduce both the early gain and the eventual coin-flip between improvement and degradation that the theory predicts, thereby closing the loop between our formal analysis and RLIF’s empirically observed non-monotone behavior.

7 Conclusions and Limitations

This work delivers a unified account of reinforcement learning dynamics in LLMs, coupling systematic empirical analysis with a rigorous theoretical framework within a single coherent explanation. Viewing RL through the lens of reasoning-pattern selection, we establish that RL reshapes the distribution over reasoning patterns through sparse, token-level interventions while leaving each pattern’s intrinsic success rate intact, and we translate this observation into provable training dynamics for both dominant reward mechanisms: for RLVR we characterize the fast-convergence and entanglement regimes and identify the base model’s reasoning quality as the decisive factor governing convergence, whereas for RLIF we derive both its early gains and its eventual degradation as two phases of one

and the same trajectory. Case studies and numerical simulations confirm every prediction, and supplementary experiments across model families and scales—from Qwen2.5-3B to QwQ-32B—show that the same mechanism governs RL training throughout. Our analysis rests on deliberate simplifications that simultaneously mark the natural directions for future work: the interpretability of LLM-identified reasoning patterns merits further validation, the two-stage framework can be extended to richer real-world reasoning settings, and the dynamics results can be generalized beyond the specific base-model assumptions adopted here. We regard these not as limitations of the account but as opportunities to broaden a framework that, in its present form, already accounts for the full range of RL training behaviors observed across our experiments.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

Section 4.2 analyzed the reasoning-pattern dynamics on Number Theory tasks from MATH. To establish that this behavior is not task-specific but reflects a general property of the two reward regimes, we repeat the dynamic analysis on Geometry tasks (Figure 3). The same contrast recurs unchanged: under RLVR the model steadily concentrates probability on the higher-accuracy patterns, whereas under RLIF pattern selection remains unstable. This replication is fully consistent with the empirical and theoretical results of the main paper and confirms that the mechanism we identify governs RL training across tasks rather than for any single one.

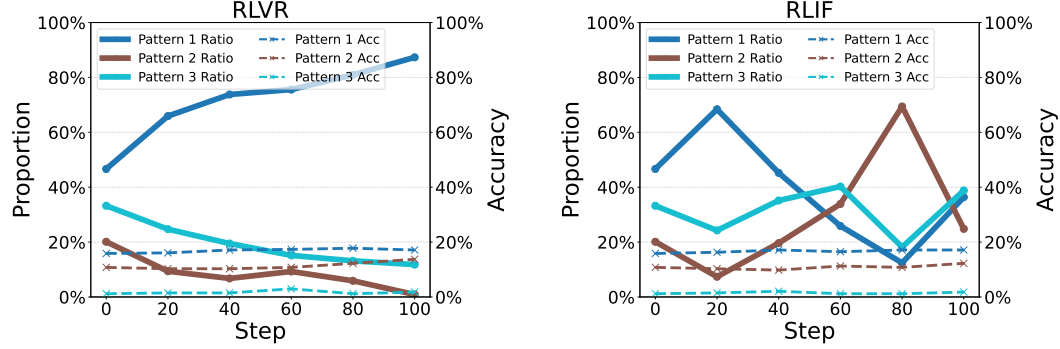


Figure 3: Reasoning-pattern distribution and corresponding per-pattern accuracy of RL-enhanced models on Geometry tasks. Under RLVR the model converges toward the highest-accuracy pattern (Pattern 1), whereas under RLIF the pattern distribution remains unstable—precisely mirroring the Number Theory dynamics of the main paper and confirming that the contrast is task-independent.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To establish that our findings hold across model families rather than being an artifact of a single base model, we conduct supplementary experiments that validate both the empirical insights and the theoretical conclusions (Figure 4). Here we take Qwen-2.5-7B-Instruct [39] as the base model and compare it against Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced with RLVR for mathematical reasoning. Our evaluation spans a broad range of mathematical domains—number theory, geometry, algebra, calculus, and counting and probability—using challenging problems sampled from the MATH dataset [16], and we extend it further to complex problems drawn from AMC23 [1]. Across every domain the results consistently reinforce our earlier findings: RLVR raises the frequency of the high-accuracy reasoning patterns while suppressing the less effective ones. Task-specific analyses are provided in Appendix B.3.

Across these tasks, the highest-success-rate patterns consistently become more prevalent after RLVR enhancement, reinforcing the findings of Section 4 and Section 5. Notably, in tasks such as Algebra and AMC23-19 (Figure 4), the per-pattern success rates remain stable across the base and RL-enhanced models; this not only echoes our earlier observations but also furnishes direct empirical support for Assumption 5.1, the foundation on which our theoretical analysis rests.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To demonstrate that our theory scales to substantially larger models, we further experiment on QwQ-32B [32], which is built on Qwen-2.5-32B-Instruct [39] and markedly strengthens reasoning through RLVR. We measure the reasoning-pattern shift between these two models on four tasks: number theory and geometry tasks from the MATH dataset [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

The result is unambiguous: the reasoning patterns of QwQ-32B predominantly coincide with the most accurate patterns of Qwen-2.5-32B, exactly as our theory predicts and in agreement with the behavior observed at smaller scale. We omit per-pattern accuracies for QwQ because API-only access

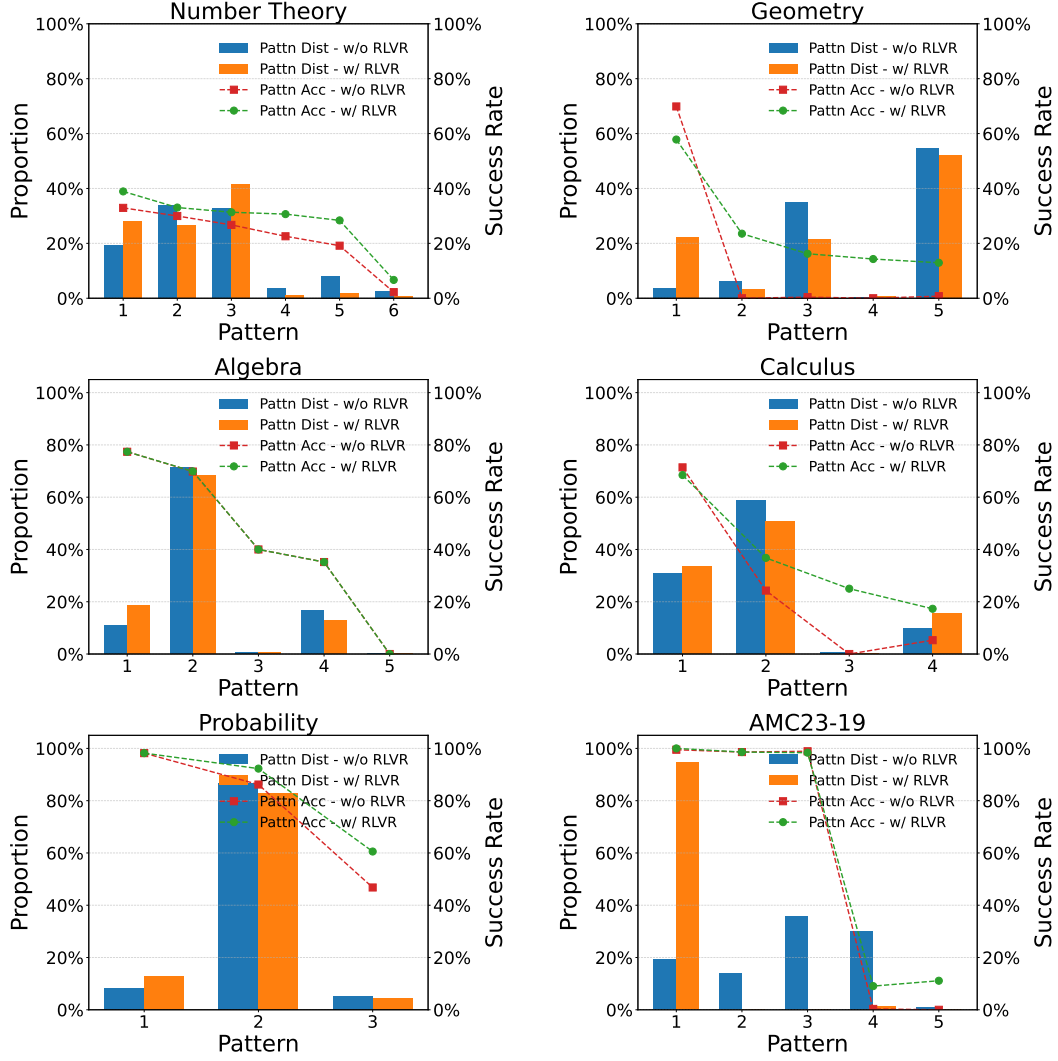


Figure 4: Reasoning-pattern distributions and corresponding success rates, with and without RLVR enhancement, across additional tasks; the results align closely with both our experimental insights and our theoretical conclusions. Bars (Pattern Dist) give each pattern’s proportion among all patterns, and dots (Pattern Acc) give its corresponding per-pattern accuracy or success rate. In every task without exception, RLVR shifts probability mass toward the highest-accuracy patterns.

precludes extracting the CoT data needed to have the base model continue from a fixed reasoning prefix—the procedure required to attribute accuracy to a reasoning pattern rather than to the model as a whole; see the “Accuracy Analysis” in Appendix B.2 for details.

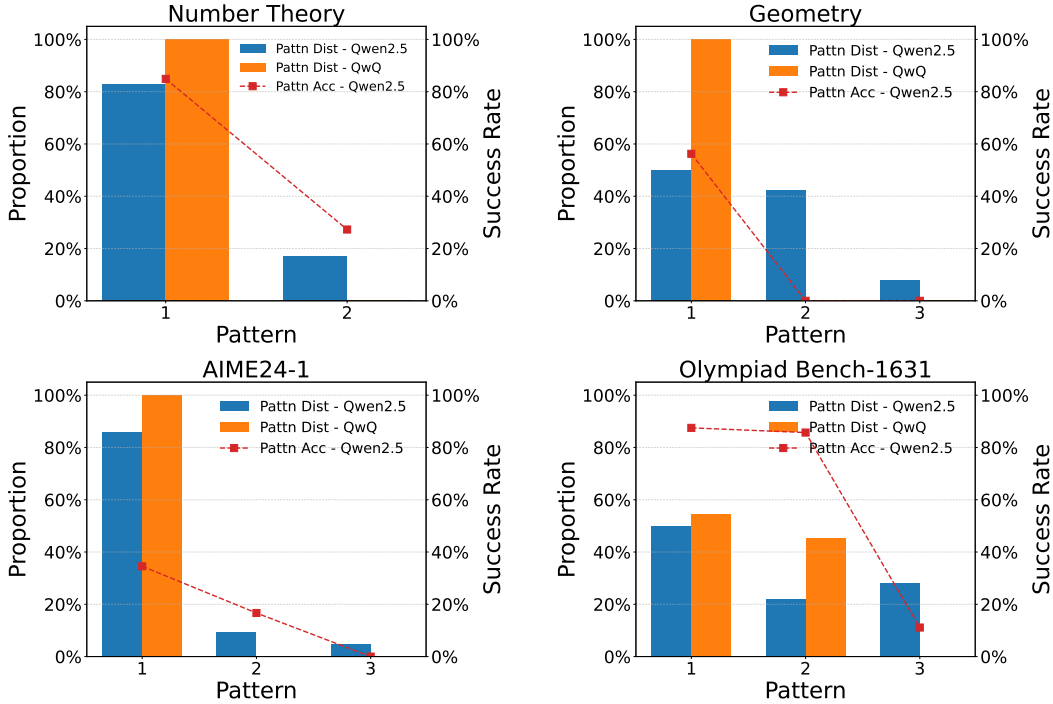


Figure 5: Evaluation on the larger models Qwen-2.5-32B and QwQ-32B. QwQ’s reasoning patterns concentrate on the most accurate pattern of Qwen-2.5, demonstrating that our theory extends cleanly from the small-scale to the large-scale regime.

B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We take Qwen2.5-3B as the base model and train it on the MATH dataset under both RLVR and RLIF. All runs are conducted within the verl [31] framework under hyperparameter settings identical to verl’s provided MATH example; RLIF differs from RLVR solely in the reward computation, which isolates the reward as the only source of variation between the two paradigms. For each method we obtain three independent training curves by varying the random seed, so that the reported contrast is provably not an artifact of any single seed.

Token-level Analysis: This experiment employs Qwen2.5-3B together with the RL-enhanced models described above. We randomly sample 400 questions from each of the GSM8K and MATH test sets and use the full AIME24 dataset. For each question we first sample an answer from the base model, then concatenate this answer to the question and feed the result to the RL-enhanced model as a new prompt. Using the OpenAI library interface, we compute, for every token in the prompt, its probability together with its rank among the probabilities of all candidate tokens at that position. Contrasting each token’s rank before and after RL yields the proportion of tokens whose rank changed—a direct measure of how sparsely RL intervenes at the token level.

Reasoning-pattern Level Analysis: We conduct two complementary experiments. First, to trace the pattern distribution *during* training, we use our own models—Qwen2.5-3B and its RL-enhanced counterpart—evaluating every 20 steps to resolve the within-training dynamics at fine temporal resolution. Second, to test the generality of our conclusions across distinct and substantially larger base models, we evaluate the open-source Qwen-2.5-7B-SimpleRL-Zoo and QwQ-32B against their

641 respective base models; for these, only the pre- versus post-RL pattern distributions are accessible, so
642 we characterize the induced distribution shift rather than the full within-training trajectory.

643 B.2 Reasoning Patterns Analysis Pipeline

644 **Model Inference:** We randomly sample questions spanning several task types (e.g., Geometry, Al-
645 gebra) and difficulty levels (Level 4, Level 5) from datasets including MATH. For the within-training
646 pattern-dynamics experiment, the model samples 1024 answers per question; for the experiment that
647 considers only the pre- versus post-RL distribution change, it samples between 64 and 1024 answers
648 per question, according to the task category. These sampled responses constitute the input to the
649 subsequent pattern-extraction and classification stages.

650 **Reasoning Patterns Extraction:** From a sampled subset of base-model responses, we use the GPT-
651 4o [19] API to summarize the pattern categories; the full extraction prompt appears in Appendix D.1.
652 To render extraction deterministic and reproducible, we fix the GPT-4o decoding temperature to 0.

653 **Responses Classification:** We again employ GPT-4o, at decoding temperature 0, to assign every
654 response to exactly one of the extracted pattern categories; the full classification prompt is given in
655 Appendix D.2.

656 **Accuracy Analysis:** To disentangle the effect of the pattern distribution on accuracy from any
657 change in the reasoning process itself, we hold the reasoning process fixed and vary only the
658 completion. Concretely, we take a base-model response, remove the sentence containing the final
659 answer, append the remaining prefix to the question, and use this as a new prompt from which the
660 RL-enhanced model continues to predict the answer. Because the partial response fixes a specific
661 reasoning pattern, the accuracy of the model’s continuation measures the success rate of that pattern
662 rather than of the model as a whole, thereby allowing us to attribute accuracy changes solely to the
663 pattern distribution.

664 B.3 Details for Each Task

665 Below we specify the experimental procedure for each task, together with task descriptions and repre-
666 sentative reasoning-pattern examples (Table 2). Example task prompts are provided in Appendix D.3.

667 **Number Theory:** This task poses coin-distribution problems over multiple bags: the bags initially
668 hold equal numbers of coins, and after additional coins are received and redistributed so that every
669 bag again holds the same count exceeding a specified total, the model must determine the minimum
670 initial number of coins per bag. We adapt this task from a level-5 MATH problem (id: test/num-
671 ber_theory/1055) [16], generating 32 variants by varying parameters such as the bag count and coin
672 totals to form the evaluation set. We adopt one-shot prompting with a simple, unrelated example that
673 fixes the answer format via "boxed" notation without biasing the reasoning approach. For evaluation,
674 we extract reasoning patterns from 20 questions with 4 responses each, sample 64 responses per
675 question, and then classify all responses according to the extracted patterns, which enables a direct
676 comparison of the reasoning-pattern distribution before and after RL.

677 **Geometry:** In the geometry task, the model is given the lengths of two sides of a scalene triangle
678 and asked how many distinct integer values (in centimeters) the third side can take. This task is
679 adapted from a level-4 MATH geometry problem (id: test/geometry/1046). Its evaluation follows
680 precisely the same pipeline as Number Theory: we construct a synthetic set of 32 questions, extract
681 reasoning patterns from 20 questions with 4 base-model (Qwen2.5-7b-Instruct) responses each, and
682 then compare the model with and without RLVR enhancement across the full evaluation set at 64
683 responses per question.

684 **Algebra:** The algebra task is a can-recycling problem: starting from a given number of old cans,
685 and given that n old cans can be recycled into one new can, the model must determine how many cans
686 can ultimately be produced. The difficulty resides in the iterative accounting—at each recycling step
687 some cans cannot be divided evenly, and these leftovers may subsequently be combined for further
688 recycling. The template is adapted from a level-4 MATH algebra problem (id: test/algebra/2768).

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)}) = 18$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Representative reasoning-pattern examples across the full suite of evaluation tasks, together with their defining keywords and common elements.

For this task and all subsequent supplementary tasks, we adopt exactly the same settings as the earlier tasks, including dataset size and number of samples.

Calculus: The calculus task is deliberately compact: the model is given a single-variable trigonometric equation together with the domain of the variable and must determine the number of solutions. Though elementary in form, the task directly probes the model’s foundational competence on calculus problems. Its prototype is a level-4 MATH pre-calculus problem (id: test/precalculus/1140).

Counting and Probability: We further evaluate the model on a counting-and-probability task involving white and black balls. Given the probability that, when two balls are drawn at random, one is black and one is white, together with the number of balls of one color, the model must compute the minimum number of balls of the other color. This task is adapted from a level-4 MATH counting-and-probability problem (id: test/counting_and_probability/79).

AMC23 Question: For this task we use question 19 of AMC23 [1]: we randomly sample 48 responses for reasoning-pattern extraction and a further 1024 responses for the subsequent pattern analysis.

B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

We evaluate four tasks. Because API access constrains testing, we select one question per task, sample 64 responses from each of the two models, and perform reasoning-pattern extraction and

classification directly on these responses. Specifically, from the MATH dataset we use question 1055 (number theory) and question 1046 (geometry), and for the remaining two tasks we use problem 1 from AIME24 and question 1631 from OlympiadBench. Representative patterns are reported in Table 2.

C Deferred Proofs

This appendix presents the complete proofs of our main theoretical results: Proposition 5.2 together with Theorems 5.3, 5.4, and 5.6.

C.1 Proof of Proposition 5.2

Proof of Proposition 5.2. In this proof, we adapt the proof techniques of [26]; recall that the optimization objective of RL is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

Under our framework, as state in section 5.1, we consider a policy conduct reasoning by first sample $\mathbf{r}_i \in \mathcal{R}$ based on $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$ and then provide the final answer $\mathbf{a} \in \mathcal{A}$ by $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$, the reward $\phi_{\text{RL}}(\theta)$ can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{r}|\mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$, then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot|\mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

Where the third equation is by Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the partition constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta}(\mathbf{r}|\mathbf{q}),$$

is a valid probability distribution such that $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$. Since Z is not a function of $\mathbf{r}$.

Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} &\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ &= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$, for RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \log |\mathcal{A}| \right)}_c, \end{aligned}$$

727 where the second term is the same for all $\mathbf{r} \in |\mathcal{R}|$, for RLIF, the optimal policy can be written as

$$\begin{aligned}\pi_{opt}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{ref}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{ref}}(\mathbf{r}_i|\mathbf{q})}\end{aligned}$$

728 So we can write $R_{RLIF}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r}))$. This concludes the proof
729 of the theorem. $\square$

730 C.2 Dynamics of Tabular Policy

731 Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the
732 gradient dynamics of θ can be computed as

$$\begin{aligned}\frac{d}{dt} \theta(t) &= \nabla \phi_{RL}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})],\end{aligned}$$

733 Where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to $\mathbf{q}$, and
734 $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned}\frac{\partial \theta_{r_i,q}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i}] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}))] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})\end{aligned}$$

735 where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we
736 have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j,q}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j,q}(t)}{\partial t}$$

737 C.3 Proof of Theorem 5.3

738 The proof of Theorem 5.3 proceeds in three steps: we first establish that the overall accuracy is non-
739 decreasing, $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4); we then derive a lower bound on $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$
740 (Eq C.5); and we finally obtain from it the convergence-time bound on t stated in the theorem.

741 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \theta_{r_i,q}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

742 For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned}\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j,q}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j,q}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}),\end{aligned}$$

743 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) p^*(\mathbf{r}_i) \\
&= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \tag{C.4}
\end{aligned}$$

744 As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern
745 $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$
746 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})) \\
&\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2 \geq 0. \tag{C.5}
\end{aligned}$$

747 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})$ holds for $t \geq 0$, define $C =$
748 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^* | \mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2.$$

749 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})}}.$$

750 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \epsilon.$$

751 This concludes the proof of the theorem. □

752 C.4 Proof of Theorem 5.4

753 The proof of Theorem 5.4 proceeds as follows: we first derive an upper bound on $\pi_{\theta(t)}(\mathbf{r}' | \mathbf{q})$ (Eq C.6),
754 then analyze the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})}$ (Eq C.7), and thereby establish that $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$
755 holds for all $t > T_0$.

756 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 757 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
 \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
 &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}') \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
 &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
 \end{aligned}$$

758 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

759 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\
 &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0
 \end{aligned}$$

760 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 761 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

762 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}
 \end{aligned}$$

763 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

764 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

765 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

766 Let $C_2 = 1/\Delta, C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

767 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

768 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

769 This concludes the proof of the theorem. $\square$

770 C.5 Proof of Theorem 5.6

771 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$, Assumption
772 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q)) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

773 where $\text{Con}_{\theta(t)} = -\sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q}))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 774 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, \mathbf{q})) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) \right). \end{aligned}$$

775 As we consider the case $|\mathcal{R}| \rightarrow +\infty$ and the success rate $p^*(\mathbf{r})$ are i.i.d. samples from $U[0, 1]$.
 776 By the Central limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ is concentrated
 777 around the its expectation 0.5, the deviation is of order is $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We define this deviation as
 778 $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$.

779 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$,
 780 the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$
 781 converges to the density: $f(x) = 1 + 12\delta(x - 0.5), x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate
 782 the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3}\delta > 0$$

783 Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the
 784 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | \mathbf{q}, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

785 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$
 786 is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

787 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

788 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 789 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 790 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 791 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- 792 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
 793 $1 - 6\delta$.
- 794 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
 795 $1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

796 This concludes the proof of the second statement.

797

□

798 D Full Prompts

799 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge
2. derivation combined with verification
3. independent calculation, suitable for learners starting from the basics,
4. detailed manual derivation, suitable for learners who need to deeply understand each step
5. etc

Or you can classify them into:

1. knowledge call type
2. derivation and verification type
3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

800

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

801

802 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

803

804 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>

user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

805

Prompt Example for Long Arithmetic

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>user

Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

806

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have

807

so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

808

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

809

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

810

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

811

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

812

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

813

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a s -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

814

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

815